TMS / SYSTEM OVERVIEW

Power and protection

4S electronics power
5 V + analog branch
Independent RF 3.3 V

File: power.kicad_sch

12–16.8 V input –> 4.981 V nominal: 1 A plan.
Teensy +3V3: ADC logic. +3V3_RF: radio only.

Thrust acquisition

LC_MOSI
LC_MISO
LC_SCK
LC_CS_N
LC_DRDY_N

Bridge + buffers + ADS1256

LC_VREF

File: thrust.kicad_sch

Ratiometric reference

LC_VREF

VEXC/2 buffer

File: reference.kicad_sch

Reference ~2.49 V
Tracks +5VA excitation.

Teensy controller

Teensy 4.1
SD logging
Split SPI + UART

LC_MOSI
LC_MISO
LC_SCK
LC_CS_N
LC_DRDY_N
P_MOSI
P_MISO
P_SCK
P_CS_N
P_DRDY_N
XB_TX
XB_RX
XB_CTS_N
XB_RTS_N
FIRE_CMD

File: controller.kicad_sch

Pressure acquisition

P_MOSI
P_MISO
P_SCK
P_CS_N
P_DRDY_N

Pressure + ADS1220

File: pressure.kicad_sch

XBee telemetry

XB_TX
XB_RX
XB_CTS_N
XB_RTS_N

XBee UART / RP–SMA

File: radio.kicad_sch

Ignition interface

FIRE_CMD

2S + ARM + relay

File: ignition.kicad_sch

TARGETS: 1 kHz per channel, SD primary: telemetry 60 Hz / 32 bytes / 115200 baud.
A0 2026–10–13: initial circuit. A1 2026–10–14: presentation and datasheet audit; electrical circuit retained.
Read the verification guide for value calculations, missing recommendations and open checks.

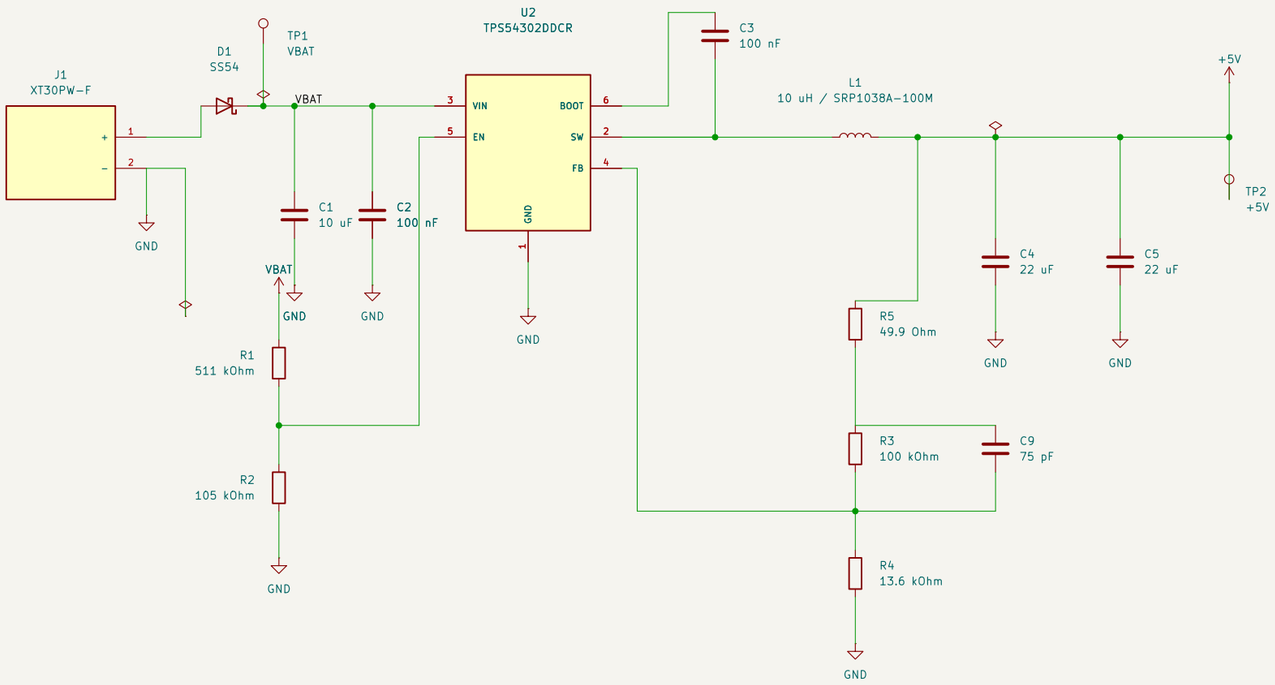
Engineering draft – see verification guide
Presentation revision; electrical circuit unchanged
TMS – stand instrumentation

Sheet: /
File: tms.kicad_sch

Title: TMS / System overview

Size: A3	Date: 2026–10–14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 1/8

POWER / 4S INPUT TO 5 V

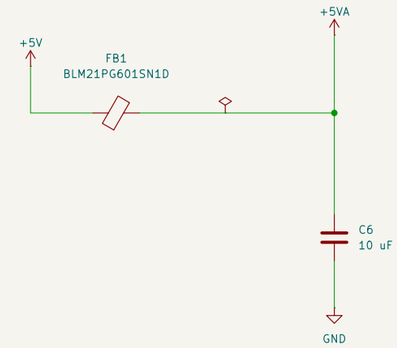


5 V SETPOINT AND LOAD

VOUT = 0.596 x (1 + 100.0499k / 13.6k) = 4.981 V.
Planning load: 1 A. IC limit is not a board current rating.
C3 is the floating bootstrap capacitor. R3/R4 set VOUT;
C9 adds phase lead. R5 permits loop-injection testing.

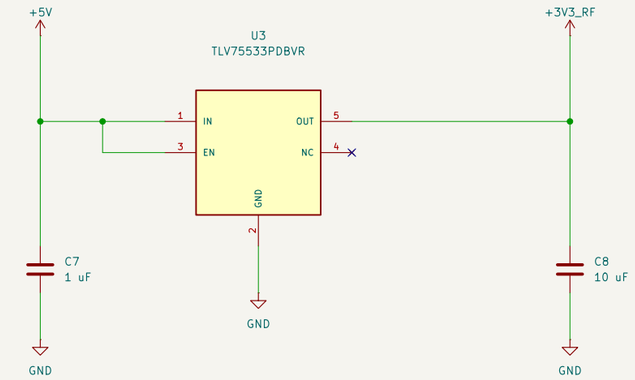
Input: 12-16.8 V. External fuse and 4S low-voltage
protection required. R1/R2 EN network is not a LiPo cutoff.
See verification guide: chapter 1 / TI TPS54302 Fig. 7-1.

ANALOG BRANCH



FB1 + C6 isolate the analog branch.
+5VA also excites the bridge (~13 mA).
Ferrite selection/noise result remains open.

RADIO SUPPLY



3.3 V / 100 mA planning load.
LDO dissipation ~0.17 W.
Keep +3V3_RF separate from Teensy +3V3.

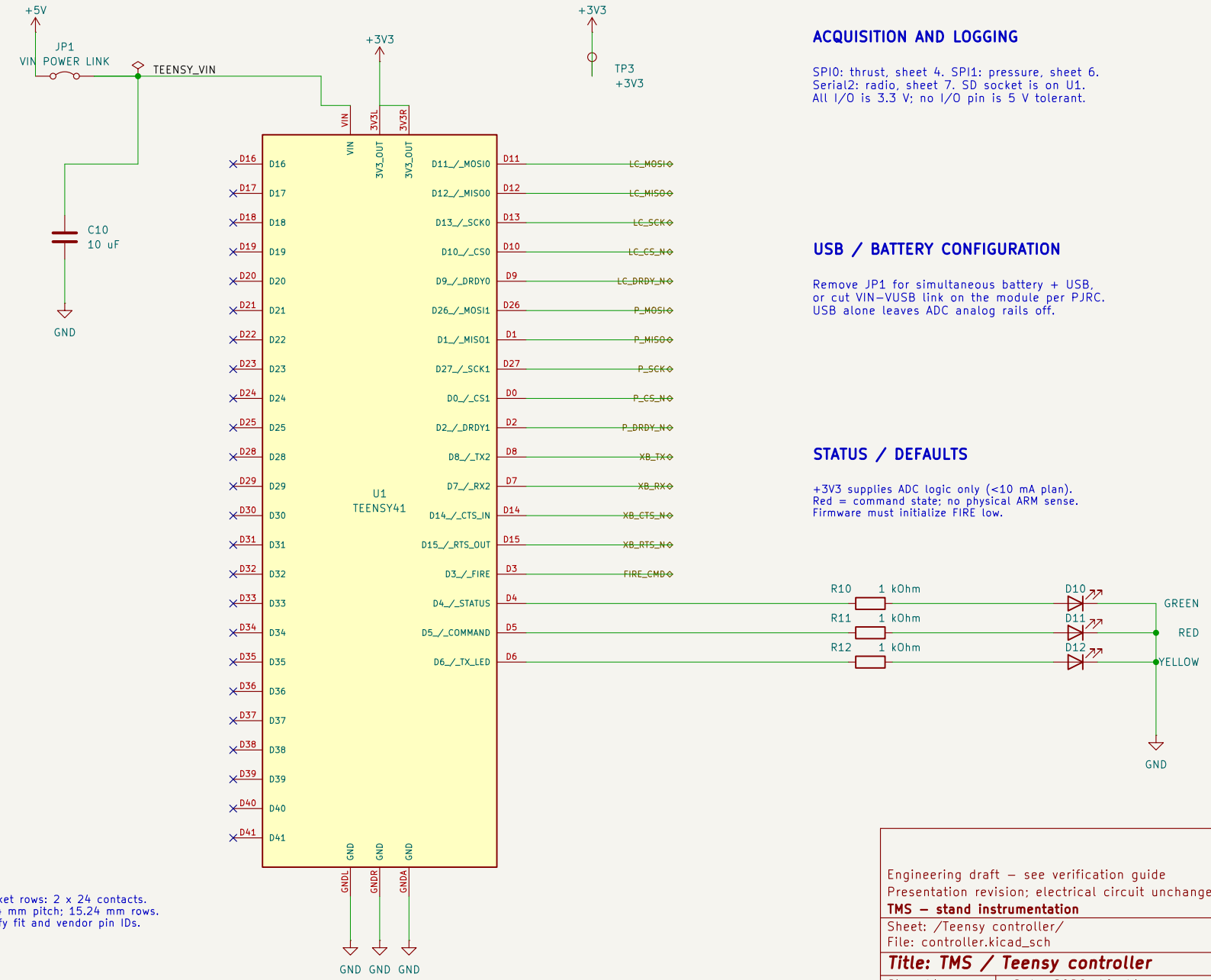
Engineering draft — see verification guide
Presentation revision; electrical circuit unchanged
TMS — stand instrumentation

Sheet: /Power and protection/
File: power.kicad_sch

Title: TMS / Power and protection

Size: A3	Date: 2026-10-14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 2/8

CONTROLLER / TEENSY 4.1



ACQUISITION AND LOGGING

SPI0: thrust, sheet 4. SPI1: pressure, sheet 6.
Serial2: radio, sheet 7. SD socket is on U1.
All I/O is 3.3 V; no I/O pin is 5 V tolerant.

USB / BATTERY CONFIGURATION

Remove JP1 for simultaneous battery + USB,
or cut VIN–VUSB link on the module per PJRC.
USB alone leaves ADC analog rails off.

STATUS / DEFAULTS

+3V3 supplies ADC logic only (<10 mA plan).
Red = command state; no physical ARM sense.
Firmware must initialize FIRE low.

Socket rows: 2 x 24 contacts.
2.54 mm pitch; 15.24 mm rows.
Verify fit and vendor pin IDs.

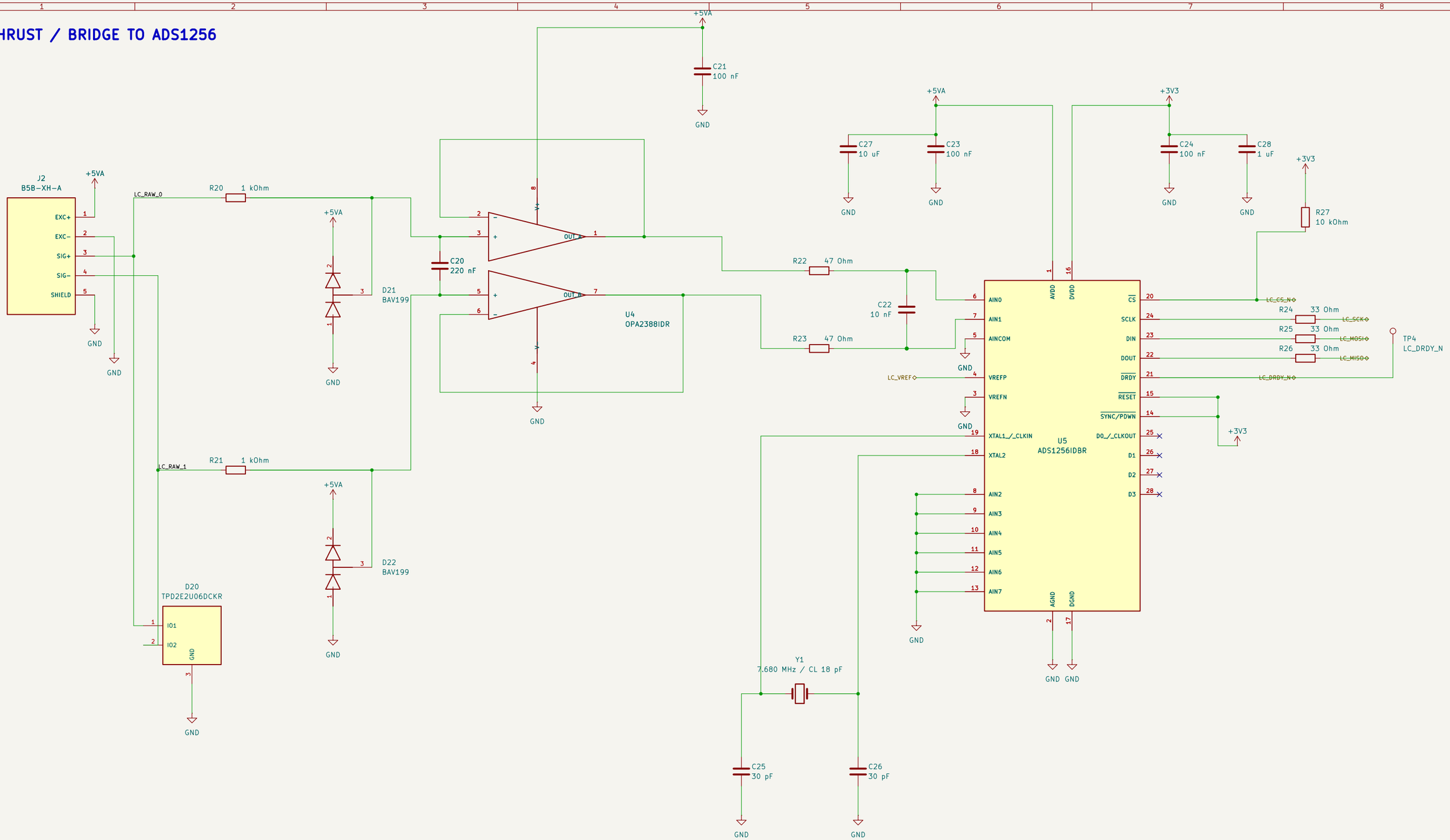
Engineering draft — see verification guide
Presentation revision; electrical circuit unchanged
TMS — stand instrumentation

Sheet: /Teensy controller/
File: controller.kicad_sch

Title: TMS / Teensy controller

Size: A4	Date: 2026-10-14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 3/8

THRUST / BRIDGE TO ADS1256



INPUT AND BUFFER

BCA-100L: nominal 10 mV at 100 kgf with 5 V excitation.
R20/R21 + C20: differential pole ~308 Hz including sensor.
U4 unity buffers avoid loading the bridge with the ADC.
Calibrate buffer offset, ADC loading and the complete chain.

ADC CONFIGURATION

AIN0-AIN1: PGA64; buffer OFF.
1000 SPS; mode 1; SCLK <=1.92 MHz.
VREF ~2.49 V; differential FS ~+/-77.8 mV.
CL estimate: 30p/2 + 3p stray = 18 pF.

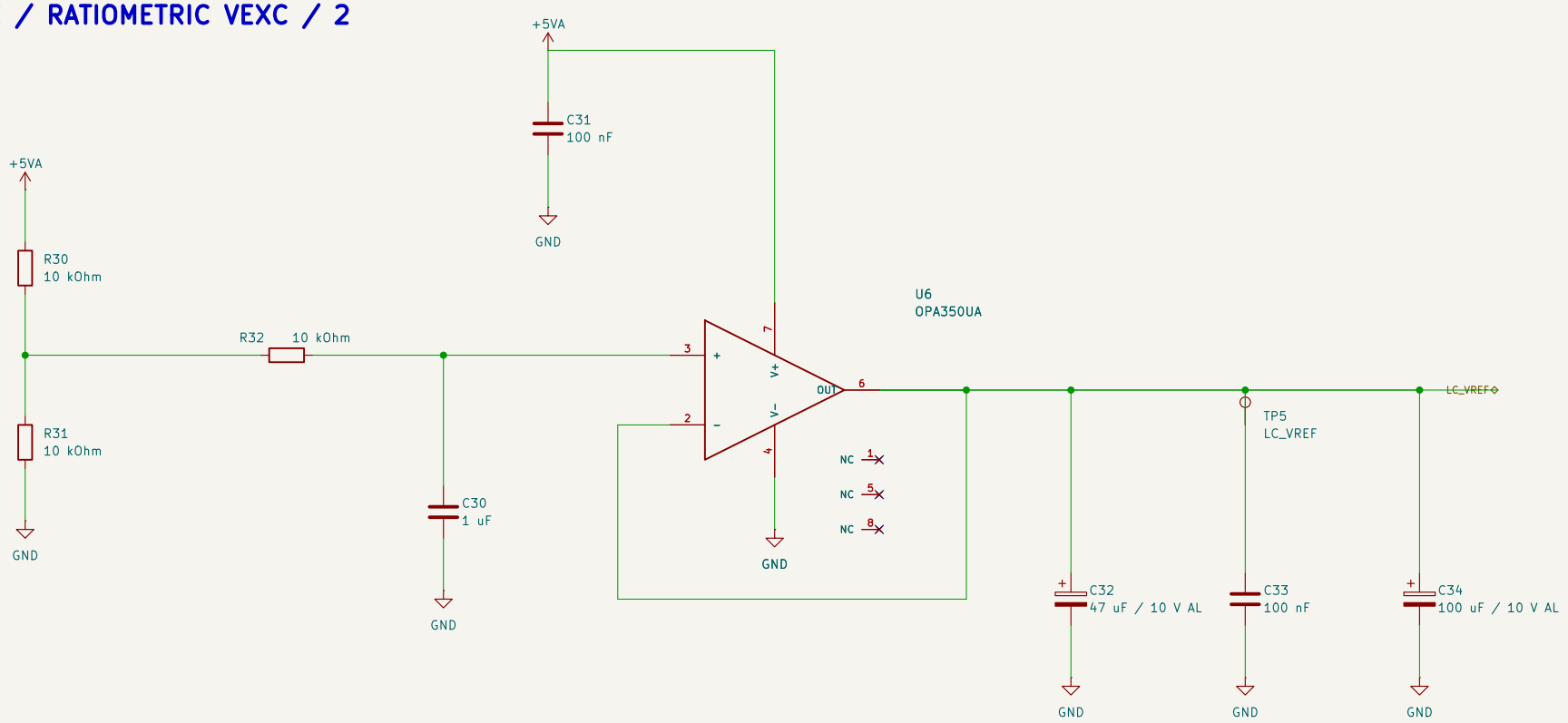
Engineering draft — see verification guide
Presentation revision; electrical circuit unchanged
TMS — stand instrumentation

Sheet: /Thrust acquisition/
File: thrust.kicad_sch

Title: TMS / Thrust acquisition

Size: A3	Date: 2026-10-14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 4/8

REFERENCE / RATIOMETRIC VEXC / 2



WHY THIS CIRCUIT

R30/R31 sense the actual bridge excitation. U6 drives ADC reference current.
VREF = VEXC x R31 / (R30 + R31) ~= VEXC/2, nominal 2.49 V.
R32/C30: (5k + 10k) x 1 uF = 15 ms; allow >=150 ms before calibration.
C32/C33/C34 follow TI Fig. 26; exact ESR, stability and settling remain open.

Place C31 at U6 and the output bank at U5 VREFP/VREFN.
Unused package pads 1/5/8 are shown NC. See guide for ratio/offset and noise checks.

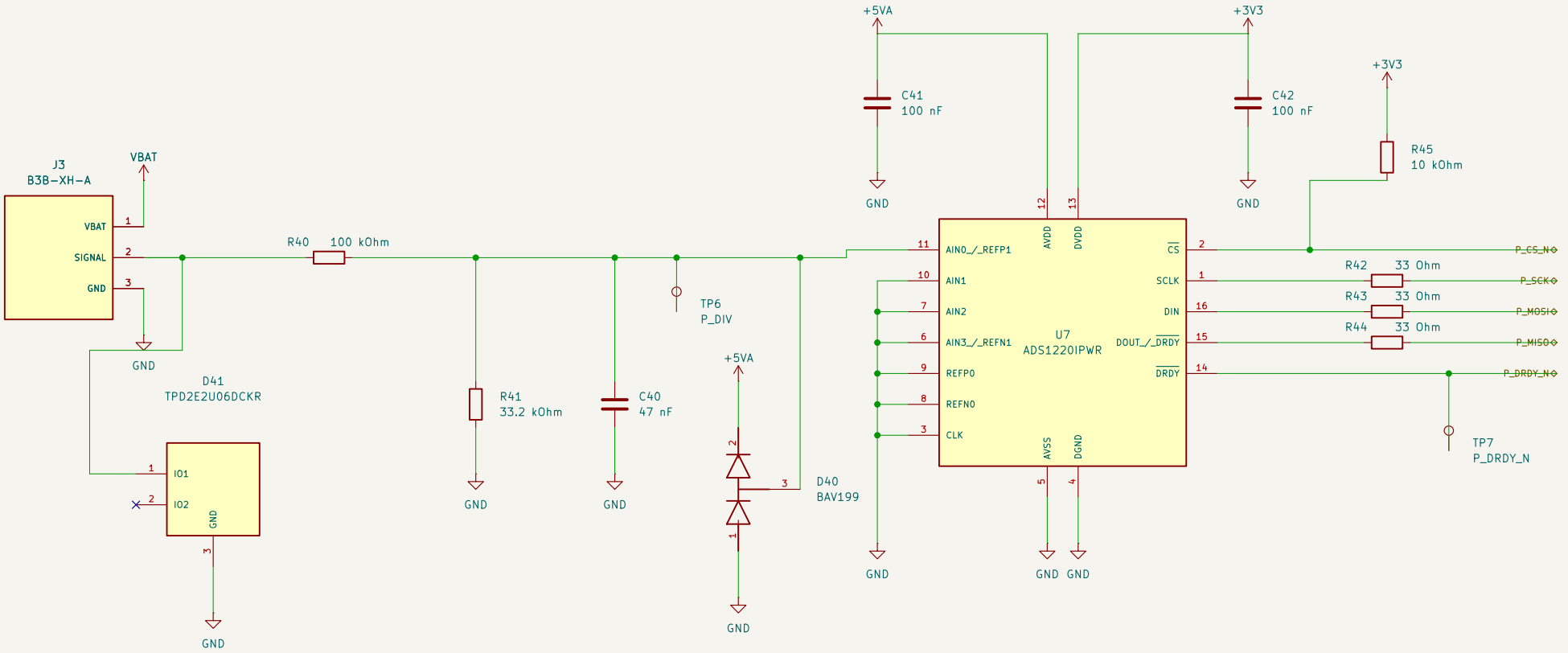
Engineering draft — see verification guide
Presentation revision; electrical circuit unchanged
TMS — stand instrumentation

Sheet: /Ratiometric reference/
File: reference.kicad_sch

Title: TMS / Ratiometric reference

Size: A4	Date: 2026-10-14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 5/8

PRESSURE / 0–5 V TO ADS1220



SCALING / INPUT FILTER

k = 33.2 / (100 + 33.2) = 0.249249; 5 V -> 1.24625 V.
RTH = 24.925 kOhm; C40 = 47 nF: fc ~136 Hz.
Sensor load 133.2 kOhm is unconfirmed; calibrate the full chain.
J3: supply / signal / return. Sensor DIN: supply / return / signal.

ADC CONFIGURATION

AIN0–to–AVSS: gain 1; PGA bypass.
Internal 2.048 V reference / clock.
Normal 1000 SPS; ~61% positive FS.
AVDD12 and DVDD13 have separate bypasses.

AUDIT: unused AIN3/REFN1 pin6 is grounded in A0; TI recommends floating.
See guide F02. 32 V sensor rating is not this board input rating; off-state faults remain open.

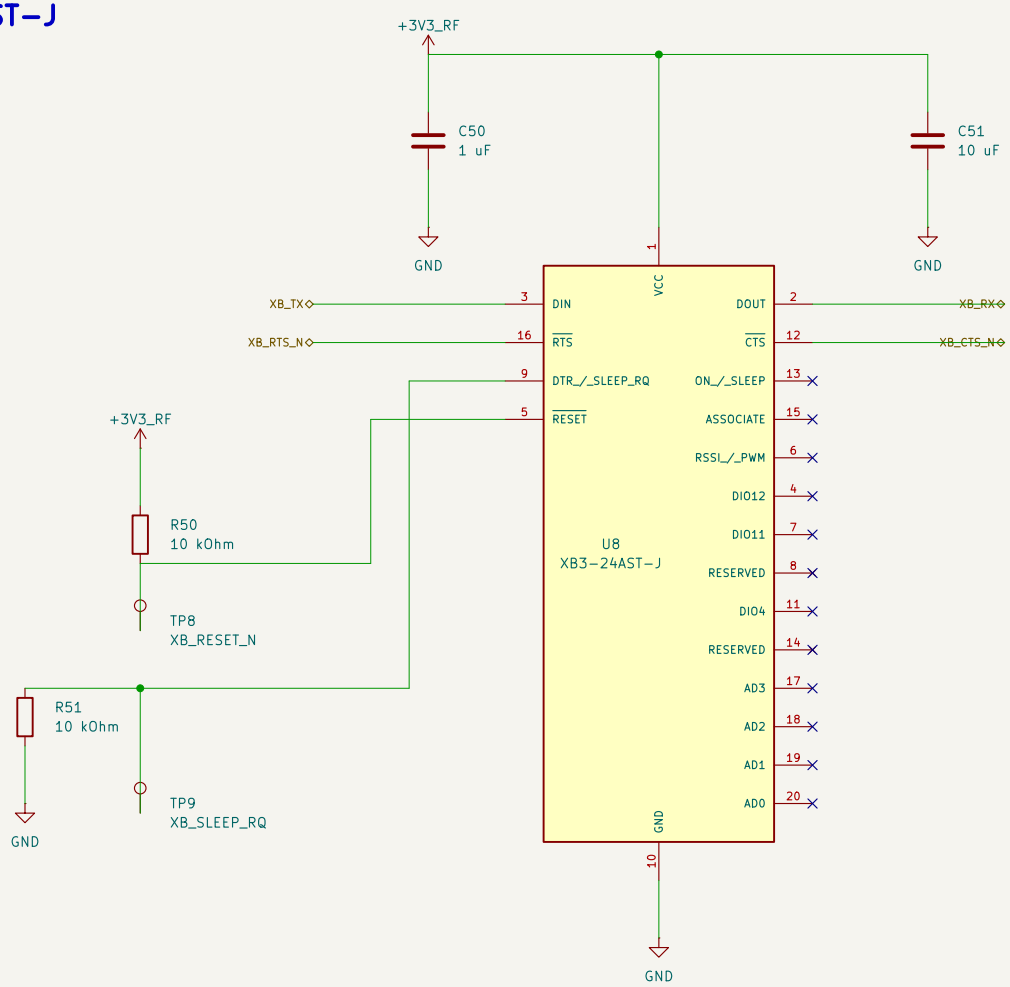
Engineering draft – see verification guide
Presentation revision; electrical circuit unchanged
TMS – stand instrumentation

Sheet: /Pressure acquisition/
File: pressure.kicad_sch

Title: TMS / Pressure acquisition

Size: A4	Date: 2026-10-14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 6/8

RADIO / XB3-24AST-J



ANTENNA / FIT

A24-HASM-450 via RP-SMA.
This variant has no U.FL.
Confirm module projection, antenna and enclosure fit.

INTERFACE / POWER

UART 115200, 3.3 V. CTS is an MCU input; RTS an output.
RF supply: independent 3.3 V / 100 mA plan. Configure DIO and sleep states.
USB adapter assumed at ground station; never join two supply outputs.

AUDIT: archived Digi manual recommends an additional 47 pF at VCC; absent.
Earlier online manual says 8.2 pF. Resolve exact revision and verify ripple; see F01/F07.

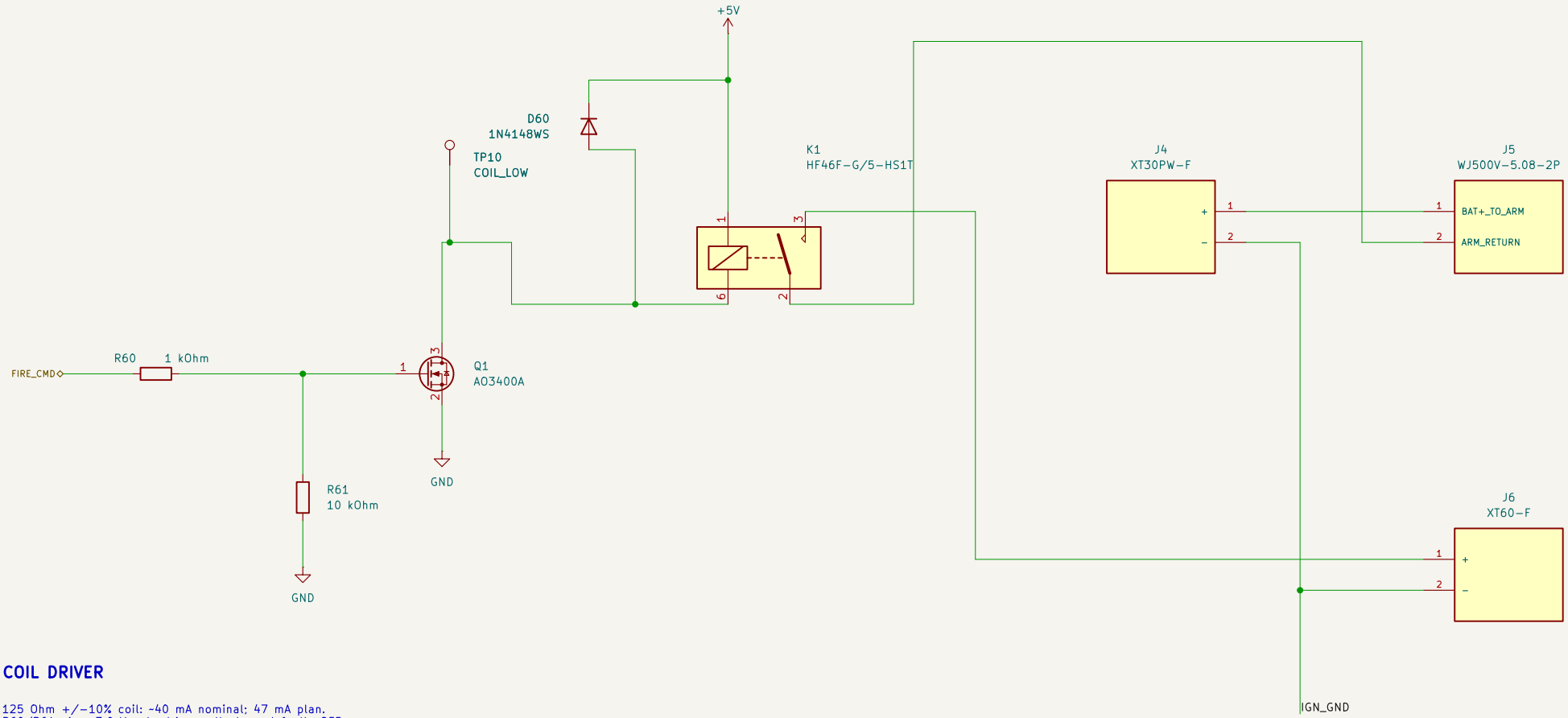
Engineering draft — see verification guide
Presentation revision; electrical circuit unchanged
TMS — stand instrumentation

Sheet: /XBee telemetry/
File: radio.kicad_sch

Title: TMS / XBee telemetry

Size: A4	Date: 2026-10-14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 7/8

IGNITION / COIL CONTROL + SERIES ARM



COIL DRIVER

125 Ohm +/−10% coil: ~40 mA nominal; 47 mA plan.
R60/R61 give ~3.0 V gate drive; pull-down defaults OFF.
D60 cathode to +5V; anode to coil drain.
Only the coil shares electronics GND.

SEPARATE CONTACT DOMAIN

2S -> physical ARM -> NO contact -> load.
IGN_GND has no electronics GND bond.
No continuity sense or physical ARM feedback.

Relay: 5 A resistive / 30 VDC. Actual load and current path remain unqualified.
Use dummy-load reset/fault tests; no live ignition approval or two-fault inhibit claim.

Engineering draft — see verification guide
Presentation revision; electrical circuit unchanged
TMS — stand instrumentation

Sheet: /Ignition interface/
File: ignition.kicad_sch

Title: TMS / Ignition interface

Size: A4	Date: 2026-10-14	**Rev: A1**
KiCad E.D.A. 10.0.5		Id: 8/8